Evaluation of multimodal Fusion for Deep Learning-Based flood extent Mapping Using Sentinel-1 and Sentinel-2

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***Abstract*—Flooding is the most frequently recorded natural disaster globally, and its increasing frequency under climate change has increased the need for fast, accurate flood extent mapping. Satellite remote sensing is crucial to this task, and Sentinel-1 Synthetic Aperture Radar and Sentinel-2 optical imagery offer complementary strengths. SAR is an active sensor that observes the surface through cloud and darkness, while optical imagery is a passive sensor that provides rich spectral detail. These two complementary features motivate combining the two through multi-sensor fusion, yet the extent to which fusion improves flood segmentation over single-sensor approaches remains unclear. This paper assesses whether a deep learning feature-fusion model improves flood extent segmentation compared to single-sensor baselines. A co-registered paired dataset, STURM-Fusion, was constructed by combining the Sentinel-2 tiles of the STURM-Flood dataset with corresponding Sentinel-1 imagery. A fusion model and two single-sensor baselines were implemented on a shared U-Net architecture with a ResNet34 encoder and compared under controlled conditions. The fusion model reached the highest overall performance, but its improvement over the optical baseline was marginal, while it outperformed the SAR baseline substantially. Per-sample and qualitative analyses showed that the fusion model behaved largely as an optical model, tending to follow whichever modality was stronger on a given sample rather than consistently exceeding both. Further analysis indicated that this outcome may be substantially shaped by the dataset, which, through its inherited preprocessing, under-represents conditions in which fusion is expected to be most valuable. The paper concludes that while fusion was the strongest approach, demonstrating its full benefit will require a dataset better suited to this task, designed to retain the conditions under which fusion is most valuable.**

I. Introduction

Flooding is the most frequently recorded natural disaster globally, accounting for 22% of all recorded disaster events in the EM-DAT international database since 1900, and causing substantial human, economic, and environmental costs to millions of people each year (Delforge et al., 2025). The frequency and severity of flooding have increased in recent years, mainly due to the effects of climate change, which intensifies extreme rainfall and rising sea levels that contribute to flood events (Tabari, 2020). Satellite remote sensing has become a central tool for flood monitoring, offering rapid, repeated observations of large, often inaccessible areas. The growing availability of satellite data has outpaced the capacity of traditional analysis methods, prompting the development of automated flood-detection approaches. Synthetic aperture radar (SAR) and optical imagery are two remote sensing techniques that are particularly suited to flood extent mapping. The Sentinel-1 (S1) and Sentinel-2 (S2) missions, operated by the European Space Agency as part of the Copernicus programme, have become established resources for flood monitoring and analysis (Munawar et al., 2022). Each mission consists of a constellation of satellites, with global coverage and regular revisit times (ESA, 2021). Their open data policy makes them valuable resources for flood extent mapping. The two sensors capture fundamentally different information. S2 is a passive optical sensor, and S1 is an active radar sensor, each with distinct strengths and weaknesses for flood mapping. SAR is an active sensor that can penetrate cloud and operate independently of daylight, allowing it to observe flooding under conditions that optical sensors cannot, while optical imagery provides rich spectral information about the surface that SAR cannot capture. These complementary traits are the central motivation for combining the two modalities. Deep learning, and convolutional neural networks (CNNs) in particular, have seen increasing adoption of deep learning for flood mapping (Bentivoglio et al., 2022). This can be attributed to their ability to learn hierarchical spatial features directly from the data. To exploit the complementary nature of SAR and optical imagery, multi-sensor fusion approaches have been proposed that combine both data sources within a single model. Multi3Net, for example, fuses imagery from multiple satellite sensors to improve flood and building-damage segmentation, demonstrating that fused models can outperform single-sensor models (Rudner et al., 2019).

This project investigates whether combining S1 SAR and S2 optical imagery through a deep learning feature-fusion approach improves flood extent segmentation compared to single-sensor models. This investigation was implemented through the following objectives: First, construct a co-registered, temporally matched paired SAR–optical dataset suitable for multimodal flood segmentation. Second, implement a feature-fusion model and two single-sensor baselines on a shared U-Net architecture. Third evaluate the three models under controlled conditions to determine whether and when fusion improves segmentation performance. The main contributions of this project are as follows. First, a co-registered paired dataset, STURM-Fusion, was constructed by combining the S2 optical tiles from the STURM-Flood dataset with corresponding S1 SAR imagery. Second, three deep learning models were constructed: two baseline single-modality models and a feature-fusion model that combines both. All models use a U-Net architecture with a ResNet-34 encoder. Finally, the models were evaluated through a combination of aggregate, per-sample, and qualitative analyses, yielding a detailed assessment of whether fusion improved performance relative to the baseline models.

II. Literature Review

Flood extent maps identify areas inundated by water during or after a flood event. Accurate flood extent mapping supports disaster response by enabling quick identification of affected regions and is essential for damage assessment and recovery planning (Bentivoglio et al., 2022). A common approach to this problem is to treat it as a pixel-wise segmentation task, where each pixel in an image is classified into predefined classes, typically flooded or non-flooded. In this context, pixel-wise segmentation is typically applied to satellite remote sensing imagery. Remote sensing is the acquisition of information about an object or geographic area without physical contact, typically using sensors that record electromagnetic energy from a distance (Jensen, 2014, p. 2-4). This provides large-scale, repeatable observations, particularly valuable for inaccessible or hazardous regions. Two well-established satellite remote sensing techniques for flood extent mapping are: multispectral optical sensors that capture surface reflectance properties (Jain et al., 2020), and synthetic aperture radar (SAR) that provides information independent of atmospheric conditions (Tavus et al., 2022). Classic approaches to the flood segmentation task have relied on manual and semi-automatic methods. Classic optical-based flood detection methods commonly use water indices such as the Normalised Difference Water Index (NDWI) (McFeeters, 1996) and the Modified NDWI (MNDWI) (Xu, 2006), which exploit differences in spectral reflectance between water and surrounding land surfaces. SAR-based flood detection methods usually rely on thresholding and change-detection techniques that exploit differences in backscatter intensity between water and non-water surfaces (Munawar et al., 2022). Classical methods rely on hand-crafted features that limit their ability to generalise across environments and scale to the large data volumes of modern satellite missions (Bentivoglio et al., 2022). These limitations have driven the development of data-driven approaches, particularly deep learning methods, to improve the accuracy and scalability of flood extent mapping. Deep learning approaches, particularly convolutional neural networks (CNNs), have become widely adopted for flood mapping. This is reflected in a rapid increase in research activity, with a considerable surge in publications occurring around 2018–2019 (Bentivoglio et al., 2022). CNNs learn hierarchical, task-relevant features directly from data without manual feature engineering, capture spatial context, and scale efficiently to large datasets (Bentivoglio et al., 2022). Most flood-mapping CNNs adopt encoder–decoder architectures for semantic segmentation, where an encoder extracts features and a decoder reconstructs a pixel-wise output (Bentivoglio et al., 2022). U-Net-based models are particularly well suited to the task, offering accurate pixel-wise classification and generalisation across geographic regions (Ghosh et al., 2024). Synthetic Aperture Radar (SAR) is an active remote sensing technique that uses microwave signals to capture surface information. The S1 mission is a constellation of two satellites (A and B) that carry C-band Synthetic Aperture Radar (SAR), enabling all-weather, day/night imaging (ESA, 2021).SAR data is well-suited for flood mapping due to its sensitivity to surface water. Smooth water surfaces produce low backscatter, resulting in a dark tone in SAR data and separating them from other land surfaces (Ghosh et al., 2024). These properties make SAR data ideal for flood mapping, however there are limitations. Radar imagery is inherently affected by speckle noise, which can reduce image quality. SAR signals can also be difficult to interpret in complex environments such as urban areas, where the double-bounce effect can cause water to produce high backscatter, leading to inconsistent identification of floodwater. Multispectral remote sensing systems are passive sensors that capture reflected electromagnetic energy across multiple spectral bands (Jensen, 2014, p. 14). Like S1, the S2 mission consists of two satellites (A and B) that provide multispectral, high-resolution observations (ESA, 2021). Multispectral imaging data is well-suited to flood mapping, as water has distinct characteristics when compared to land. Deep learning approaches have significantly improved flood-mapping performance; however, as previously stated, each modality has its limitations. While recent studies have explored the use of multimodal approaches, there remains a limited understanding of how combining SAR and optical data affects segmentation performance and model generalisation (Rudner et al., 2019). Additionally, the effectiveness of fusion strategies in improving segmentation performance remains an open area of research. Therefore, this project aims to investigate the use of multimodal fusion for flood extent mapping and to evaluate its ability to improve segmentation accuracy in comparison to single-modality approaches.

III. Methodology

A. STURM-Fusion Dataset

This project uses a newly constructed paired dataset, called STURM-Fusion. The purpose of this dataset is to provide spatially matched S1 SAR and S2 optical imagery, acquired within acceptable temporal windows, for multimodal flood extent mapping. Each sample contains a Copernicus EMS-derived flood mask, a S1 SAR image acquired within 24 hours of the mask timestamp, and a S2 optical image acquired within 72 hours of the mask timestamp. STURM-Fusion is derived from the STURM-Flood dataset. STURM-Flood provides both

S1 and S2 satellite image tiles of size 128 × 128 pixels at 10 m spatial resolution, alongside corresponding Copernicus EMS-derived flood masks. While STURM-Flood includes both SAR and optical imagery, these images are not directly paired in a way suitable for fusion, as they are not spatially or temporally aligned for each sample. For this reason, the pre-processed S2 images and their corresponding Copernicus EMS-derived flood masks were used as the anchor data. Temporally and spatially matched S1 SAR imagery was then retrieved separately using Google Earth Engine, allowing each final sample to contain aligned S1, S2, and flood-mask components. The S2 Images from STURM-Flood were used as a starting point when constructing the STURM-Fusion dataset, as suitable S2 imagery is less consistently available than S1 SAR imagery due to cloud cover and other atmospheric conditions. S2 preprocessing is also comparatively more complex. For this reason, the pre-processed S2 tiles and matching EMS masks from STURM-Flood provided the most reliable anchor data. The first stage of construction involved selecting S2 samples where the acquisition time was within 72 hours of the corresponding flood mask timestamp. This threshold was selected to reflect the reduced availability of suitable S2 optical imagery, balancing temporal alignment with the flood mask against the need to retain enough usable samples. For each selected S2 tile, the tile bounds and geospatial information were extracted from the image metadata. These bounds defined the area of interest for searching for matching S1 imagery. To retrieve the S1 tiles, Google Earth Engine provides direct access to the S1 image archive. It supports geospatial and temporal filtering, image processing, and export within a single cloud-based platform (Gorelick et al., 2017). Candidate images were filtered to within 24 hours of the flood mask timestamp. A shorter temporal window was used for S1 to leverage the wider availability of usable SAR imagery. Subsequent verification of the S1 images was performed to ensure that each image covered the same 128 × 128 pixel area as the corresponding S2 tile and flood mask. A small tolerance was allowed for incomplete coverage, with samples retained only where no more than 5% of pixels contained no-data values. The final paired samples were then compiled into the new STURM-Fusion dataset. The Sturm-Flood flood maps were derived from Copernicus EMS mapping products, reprojected to the appropriate UTM zone, rasterised to a 10 m pixel size to match the Sentinel imagery, and clipped to the relevant area of interest. The original STURM-Flood masks classify water-related features into multiple classes, including flooded areas, river-related features, open water, reservoirs, lakes, coastline/AOI, and nodata regions (Notarangelo et al., 2025). For this task, the multiclass masks were converted to a binary class mask. The classes were simplified to only distinguish between water/flood and non-water/background pixels. Water-related classes were mapped to the positive class, while non-water/background pixels were mapped to the negative class. The S2 optical images used in STURM-Fusion were taken directly from the STURM-Flood dataset. In STURM-Flood, S2 preprocessing included cloud and shadow masking, selection of relevant visible, near-infrared, red-edge, and shortwave infrared bands, and sharpening of lower-resolution bands to a common 10 m spatial resolution. The resulting imagery consists of 9-band, 128 × 128-pixel optical tiles (Notarangelo et al., 2025). The S1 SAR images used in STURM-Fusion were retrieved from Google Earth Engine using the geospatial metadata extracted from the corresponding S2 tiles. The preprocessing steps were selected to make the retrieved Google Earth Engine imagery as consistent as possible with the S1 data used in STURM-Flood. Preprocessing includes images to ensure only the VV and VH polarisation bands are present. Images were cropped where necessary to ensure a fixed size of 128 × 128 pixels. No-data values were converted to zero to avoid errors in statistical calculations. Bands were then clipped using fixed per-band value ranges, VV: -30 to 5 dB, VH: -35 to 0 dB. Following clipping, each band was normalised independently using its own mean and standard deviation. A Lee speckle filter was applied separately to each SAR band to reduce speckle noise while preserving local spatial structure (Lee, 1980).

B. *Dataset Composition and Split Strategy*

A key challenge in using STURM-Fusion was its uneven composition and limited size. The data is highly imbalanced, with most tiles containing only a small proportion of flooded pixels. At the event level, some flood events contributed many more samples than others(per-event sample counts, flood coverage, and class composition are shown in Appendix A). There is substantial variation in the distribution of samples: two events, EMSR470 and EMSR479, dominate the dataset, while three of the smallest events have sample sizes ranging from 2 to 12. This imbalance is important because each flood event may contain different features such as land-cover types, flood characteristics and sensor conditions. If a small number of large events dominate the training data, the features learned from them could adversely affect a model's performance on unseen data. This imbalance can affect model performance as it may achieve reasonable performance by predicting background pixels more frequently, while still failing to detect smaller flooded regions. For this reason, metrics such as IoU, F1-score, precision, and recall were considered more appropriate for evaluating flood segmentation performance. Flood coverage also varied considerably between flood events. Some events contained tiles with high proportions of flooded pixels, while low-flood-coverage samples dominated others. Event-based splitting was initially explored as it provides a stricter test of generalisation to unseen flood events. However, the limitations of the dataset led to unstable model behaviour across experiments. Although the models regularly learned the training data, validation performance varied considerably. Given the dataset's composition, no event-based ordering could fully resolve these issues. This led to the adoption of a random split strategy. Random splitting provided a more stable and controlled experimental setup by distributing samples from different flood events across the training, validation, and test sets. Although this didn't solve the imbalance issues, it helped mitigate them, as the more dominant events and classes were more evenly represented in the dataset. As a result, training behaviour became more stable, and validation performance gave a clearer indication of model learning. This made it easier to monitor model improvement during training and enabled a more controlled comparison between the S1, S2, and fusion models. Although a random split strategy was the appropriate choice given the dataset's limitations, it introduces an important limitation. Because samples from the same flood event can appear across the training, validation, and test sets, the evaluation cannot be considered a strict test of generalisation to unseen flood events. For this reason, the results should be interpreted only as a controlled model comparison.

C. Model Architecture

A U-net architecture (Ronneberger et al., 2015) with a ResNet34 encoder (He et al., 2016) was selected as the baseline for the segmentation tasks. The same architecture was used for both S1 and S2 models to ensure a controlled comparison. The selection of U-Net was informed by its frequent use in the flood segmentation literature, where encoder–decoder architectures are commonly applied to pixel-wise classification tasks, and particularly by its use as the baseline architecture in the STURM-Flood study (Notarangelo et al. (2025); Tavus et al. (2022); Ghosh et al. (2024)). The choice of this architecture was informed by the work of (Ghosh et al., 2024), who compared several U-Net based architectures for S1 flood segmentation, including U-Net with ResNet34, InceptionV3, and EfficientNet-B7 encoders, as well as UNet++ with EfficientNet-B7. The results of this research showed that using a more complex encoder did not substantially improve flood segmentation performance. A feature-fusion model was developed to evaluate the combination of S1 SAR and S2 optical imagery following the multi-encoder, single-decoder design used for multi-modal MRI segmentation in MMFNet (Chen et al., 2019). To ensure a controlled comparison, the model retained the same U-Net ResNet34 architecture as the baseline models. The architecture of the fusion model's parallel encoder is illustrated in Figure 1, which shows the first two encoder levels. The complete architecture, including the remaining levels and the shared decoder, is provided in Appendix B. The model used two separate ResNet34 encoder streams: The S1 encoder received the two SAR channels, VV and VH. The S2 encoder received the 9 optical bands. Using separate encoders allowed each modality to be processed independently during feature extraction. To pass information from both encoders into a single decoder, the learned S1 and S2 feature representations were fused, producing a single combined representation at each encoder level. This is achieved by concatenating the corresponding feature maps, then applying a 1×1 convolution to reduce the channel count back to the expected U-Net decoder size. This method allows the two parallel encoder streams to share a U-Net decoder. The decoder progressively up-samples the fused feature maps, starting with the deepest layer and progressing back towards the original image resolution. At each stage, the upsampled feature map is concatenated with the corresponding fused skip connection from the encoder and then convolved to refine the combined features. The segmentation head then converted the final decoder output into a single-channel prediction map where each pixel represented the model's flood/non-flood prediction.

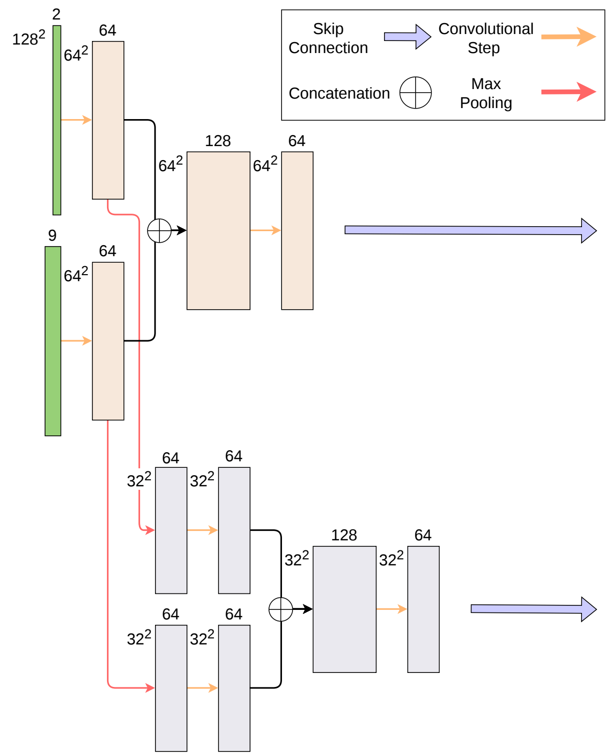


Fig. 1: Feature-fusion parallel encoder (first two levels).

D. Training Procedure

A training procedure was designed to ensure that all models were trained and evaluated under consistent conditions. The dataset was split into training, validation, and test sets using a random split with a fixed random seed; this ensures the same split is used for each model. Rotation augmentation was explored to address class imbalance, as previous flood segmentation work has used rotated tiles for this purpose (Ghosh et al., 2024). However, it was excluded from the final experiments as it negatively affected validation performance. A hybrid loss function combining binary cross-entropy and Dice loss was used during training. This was selected to address the dataset's class imbalance. Binary cross-entropy treats the task as a per-pixel binary classification problem, providing a clear learning signal for each pixel. However, BCE alone can be biased towards the dominant background class. To address this, Dice loss was combined with BCE, which compares the predicted flood mask with the ground-truth mask across whole regions. This encourages the model to increase the overlap between predicted and true flood areas. Models were trained using the AdamW optimiser (Loshchilov and Hutter, 2019), and a scheduler was used to reduce the learning rate when validation loss stopped improving. This allows the model to continue training with smaller updates when progress plateaus. The main hyperparameters considered were learning rate, batch size, weight decay, and dropout rate. A grid search was used to evaluate different combinations of these hyperparameters, enabling a consistent search across all models. During training, validation loss, IoU, F1-score, precision, recall, and accuracy were monitored after each epoch. Validation loss was used to guide the learning rate scheduler, and the best model checkpoint was selected based on validation IoU, with F1-score as a secondary criterion. Early stopping was implemented to halt training when validation IoU performance stopped improving. This was set to a patience of 8 epochs.

E. Evaluation Metrics

Model performance was assessed using two complementary evaluation methods. The first provides an overall performance measure for model selection and comparison, while the second supports a detailed analysis of how performance varies across individual samples. The primary evaluation was based on a set of metrics calculated consistently across both validation and testing. Pixel counts were pooled within each batch to produce a micro-averaged score per batch, and the mean of these per-batch scores was reported. Applying this identical procedure to validation and test data ensures that the validation performance used to guide model selection is directly comparable to the final test performance. Multiple evaluation metrics were selected, as accuracy alone can be misleading when dealing with a class-imbalanced dataset. Accuracy measures the proportion of correctly classified pixels, but most pixels belong to the non-flood class. Precision measures how many pixels predicted as flood were flood. This is useful for assessing whether the model over-predicts flooded areas. Recall measures the fraction of true flood pixels correctly detected, a good indicator of whether the model is detecting flooded areas. F1-score combines precision and recall into a single metric, making it useful for assessing how well the model manages both overprediction and missed flood pixels. F1-score was treated as a secondary metric. Intersection over Union (IoU) measures the overlap between the predicted and ground-truth flood masks. It was used as the key segmentation metric because it directly evaluates how well the predicted flood extent matches the reference flood extent. All evaluation metrics require the model's continuous probability outputs to be converted into a binary flood/no-flood prediction. A default decision threshold of 0.5 was applied, so that any pixel assigned a flood probability of 0.5 or greater was classified as flood. Because the batch size was fixed and the loaders were not shuffled, batch composition was constant across epochs, configurations and models. Batch averaging, therefore, introduces a consistent bias to absolute values but does not affect relative comparisons. To support a deeper analysis of model behaviour, all metrics were calculated per-sample, computing a separate score for each test tile rather than pooling across a batch. These per-sample values were used to examine the distribution of scores, the relationship between performance and flood coverage, and variation between flood events.

F. Implementation

The implementation was designed to support reproducibility across all experiments. The project was implemented in Python using PyTorch, with experiments run in Google Colab using GPU acceleration. Each experiment was controlled by a saved configuration file that specified the model type, data modality, hyperparameters, split method, and training settings. A fixed random seed was used to ensure that the same random train, validation, and test split was applied across all models. For each experiment, configuration files, model checkpoints, and training, validation, and test metrics were saved in the experiment directory.

IV. Results

Following the grid search procedure described in the methodology, the best-performing configuration for each model was selected by validation IoU. The selected configurations are summarised in Table I. The grid search results revealed several consistent patterns across the models. A learning rate of 0.001 and a weight decay of 1e-05 yielded the best validation performance across all three cases. The optimal batch size and dropout rate, however, differed between models. The fusion and S2 models both benefited from a larger batch size of 64 and dropout (0.2), whereas the S1 model performed best with a smaller batch size of 32 and no dropout. The fusion model achieved the highest validation IoU (0.807) among all configurations, marginally ahead of the best S2 model (0.785), while the S1 model trailed considerably behind (0.656). Training curves (loss, IoU, F1, precision, and recall) were examined to assess learning stability and generalisation (Appendix C-B). The fusion and S2 models trained almost identically: loss fell sharply and plateaued around 0.17–0.18, validation IoU settled at roughly 0.78–0.80 and 0.76–0.78, respectively, and precision and recall stayed closely balanced. The close agreement between the training and validation curves indicates that neither model overfits, and the selected checkpoints occurred at the later epochs 28 and 26. In contrast, the S1 model displayed significantly less stable training behaviour. Its loss decreased more gradually and plateaued at approximately 0.30. This instability was clearly visible across all three metric curves. This behaviour suggests that the SAR-only model struggled to extract a consistent learning signal from the two-band input.

TABLE I: Selected hyperparameter configurations and validation performance for each model.

Model	**Learning Rate**	**Batch Size**	**Weight Decay**	**Dropout**	**Val IoU**	**Val F1**
Fusion	0.001	64	1e-05	0.2	0.807	0.893
S2 (Optical)	0.001	64	1e-05	0.2	0.785	0.880
S1 (SAR)	0.001	32	1e-05	0.0	0.656	0.790

Table II reports the aggregate test set performance of the fusion model and the S2 (optical) and S1 (SAR) baselines at the 0.5 threshold, across all five metrics. On IoU, the primary segmentation metric, fusion scored 0.759, S2 0.756, and S1 0.657, placing fusion +0.003 above the optical baseline and +0.099 above the SAR baseline. The same ordering appeared in F1, with 0.859 for fusion, 0.858 for S2, and 0.790 for S1. Precision and recall followed a similar pattern, with fusion and S2 closely matched on both and S1 lower. Accuracy was high across all three models, ranging from 0.867 for S1 to 0.911 for fusion, with the optical and fusion models differing by only 0.001. Table II also reports the per-sample IoU distribution for each model, summarising the IoU values computed for individual test tiles. The mean per-sample IoU was 0.652 for fusion, 0.618 for S2, and 0.544 for S1 in each case the median exceeded the mean (0.775, 0.750, and 0.633, respectively), with a large standard deviation. The full hyperparameter tuning results and training curves for all models are provided in Appendix C.

TABLE II: Test set performance metrics for each model.

Model	Accuracy	Precision	Recall	F1	IoU	Mean IoU per sample $\pm$ SD
Fusion	0.911	0.841	0.886	0.859	0.759	0.652$\pm$0.320
S2 (Optical)	0.910	0.835	0.888	0.858	0.756	0.618$\pm$0.332
S1 (SAR)	0.867	0.814	0.770	0.790	0.657	0.544$\pm$0.317

V. Evaluation and Discussion

A. Performance and Flood Coverage

Per-sample IoU was strongly bimodal across all three models. A large concentration of tiles achieved high IoU, clustered toward 1.0, while a separate group of tiles scored at or near 0, with comparatively few tiles falling in between. This indicates that the models tend to either produce an accurate segmentation of the tile or fail the task entirely. In every case, the median IoU was notably higher than the mean, showing that near-zero failures pull the mean below the level at which most tiles perform. The standard deviation was large across all three models, indicating that performance was highly variable from one tile to the next. The fusion and S2 distributions were similar in shape, whereas the S1 distribution showed a larger share of tiles in the low-IoU region and a lower median. To investigate the clusters of clear success and failure cases, a per-sample IoU was plotted against the proportion of each tile covered by flood (Figure 2). Full per-model IoU distributions and scatter plots are provided in Appendix D. A clear relationship was evident across all three models: IoU increased with flood coverage. Tiles with a high percentage of flooded pixels consistently produced accurate flood maps. Whereas the near-zero failures were concentrated almost entirely among tiles with very low flood coverage. Below 20% flood coverage performance was highly variable and frequently collapsed. This indicates that the earlier bimodal distribution is largely driven by flood coverage, with small-flood tiles accounting for the majority of failures. Comparing the models, the fusion and S2 tracked one another closely, whereas the S1 model showed greater scatter and a larger number of low-coverage failures. Performance also varied between flood events, consistent with the dataset's uneven event composition; per-event mean IoU for each model is shown in Appendix D-A.

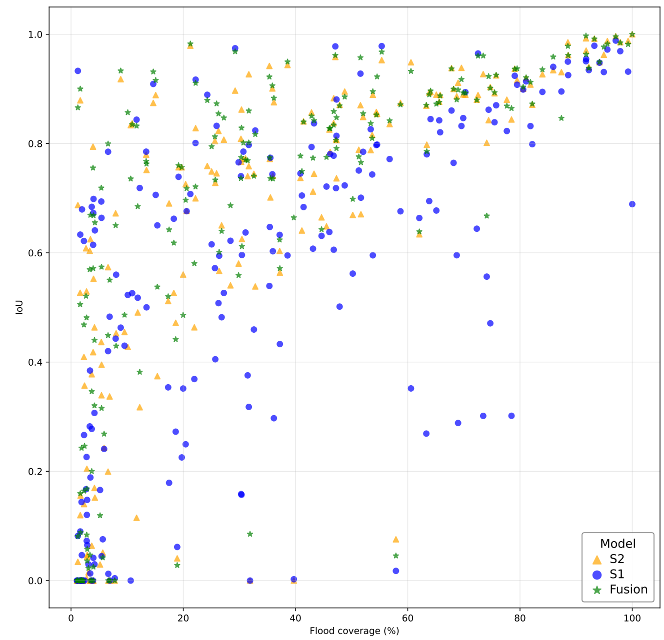


Fig. 2: Scatter Diagram of IoU and Flood Coverage

B. Qualitative Analysis of Representative Cases

1) EMSR470_AOI01_46_07_2_1: This sample is from EMSR470, a flash flood event in the Oti River region of Togo. This event affected a large rural and peri-urban area, and the Copernicus EMS product was derived from post-event satellite imagery using a semi-automatic approach (Copernicus EMS, 2020a). This sample was selected as a clear success case because the mask contained a relatively even flood/non-flood split and a distinct flood boundary (Figure 3a and Figure 3b). All three models performed well on this high-coverage tile, but their performance was not equal. Although the sample was selected as a clear case based on its distinct flood boundary and balanced flood/non-flood split, the models diverged more than expected. The S1 model achieved the highest IoU (0.974), with the fusion model close behind (0.968), while the S2 model scored notably lower (0.897). The most significant difference in performance between the models is precision, with the S2 model's precision (0.901) well below that of the S1 (0.985) and fusion (0.976) models. This indicates that the optical model produced more false positives, over-predicting the flood extent relative to the other two models. Inspection of the inputs offers a possible explanation for the optical model's weaker performance. The S2 optical image contains visible artefacts. This may be due to residual cloud or other optical noise that was not fully removed during preprocessing. The optical model's false positives appear to align with this region, suggesting that this could have been a contributing factor to its over-prediction. Notably, the fusion model's performance closely matched that of the S1 model rather than the weaker optical model in this instance (Figure 3c and Figure 3e). This suggests the complementary behaviour that fusion is intended to provide, in which the strengths of one modality compensate for the weaknesses of the other. However, this observation is based on a single case and does not establish this behaviour as a consistent characteristic of the model.

2) EMSR441_AOI05_2_3_2_2: This sample is from EMSR441, a riverine flood event in Finnish Lapland caused by spring snowmelt (Copernicus EMS, 2020b). It was selected as a difficult case because the flood region occupied only a small proportion of the tile and came from an event with relatively few samples (Figure 4a and Figure 4b). All three models failed almost completely on this low-coverage tile, each producing an IoU of 0. The ground truth was a thin, narrow riverine flood, and none of the models detected it. In each case, the predicted masks missed the flooded channel entirely (Figure 4c, Figure 4d and Figure 4e). In addition, all three models predicted a small false-positive flood region in the upper centre of the tile, where no flood was present. This case is informative precisely because the failure was shared across all modalities. The reason for this appears to be the small extent of the flood and the narrow-flooded channel, only a few pixels wide. As there are very few flooded pixels, a model that fails to capture its precise shape is penalised heavily under IoU. The shared false-positive region suggests that all three models were responding to similar misleading features in the inputs and struggled to distinguish the subtly flooded feature from other elements of the scene. Unlike the previous example, all inputs were insufficient, and fusion inherited the limitation common to both. This indicates that fusion does not overcome the difficulty both baselines have with predicting low flood presence within a tile.

3) EMSR570_AOI02_07_03_2_1: This sample is from EMSR570, a riverine flood event in Lismore, Australia, associated with prolonged rainfall during a La Niña event (Copernicus EMS, 2022). It was selected as an edge case because the tile had a more balanced flood/non-flood split but showed an irregular and visually complex boundary (Figure 5a and Figure 5b). This produced the clearest ordering of the three models, with the fusion model achieving the highest IoU (0.698), ahead of the S2 model (0.669) and the S1 model (0.562). It is the one case among the three examples in which fusion outperformed both baselines outright, rather than matching the stronger of the two. Across all three models, recall was lower than precision, indicating that the models tended to miss flooded pixels rather than overpredicting them. This can be seen in the FN/FP map where the errors are concentrated as false negatives along the complex edges of the flood (Figure 5c, Figure 5d and Figure 5e). The models seem to capture the broad flooded regions but lose accuracy along the detailed margins. The ordering of the models in this case is consistent with the overall results. The S1 model was weakest while the S2 model performed substantially better and fusion produced a modest improvement over the optical baseline. Further representative examples are provided in Appendix D-D.

C. *Cross-Model Fusion Improvement Distribution*

This analysis compares the models individually to assess the value of fusion directly at the sample level. For each test tile, the higher of the S1 and S2 IoU was subtracted from the fusion IoU, giving a positive value where fusion beat the stronger baseline and a negative value where it did not. Across the test set, these differences were sharply concentrated around zero(see Appendix D-E for the full distribution). A large majority of tiles showed little or no difference between fusion and the best single-modality baseline, with slightly more tiles falling just above zero than below and very few samples at the extremes in either direction. This indicates that, for most tiles, fusion neither meaningfully improved nor hindered the performance of the better single-modality. Rather than providing a consistent per-sample gain, fusion most often matched the best available baseline. This finding also contextualises the qualitative examples. The success case, in which fusion tracked the stronger modality and the failure case, in which fusion offered no advantage, both fall within this central mass. The edge case, an instance where fusion improved slightly over a sample that both baselines already handled reasonably well, illustrates the kind of small positive difference that, accumulated across similar tiles, may account for the slight positive lean.

D. *Legal, Social, Ethical and Sustainability Considerations*

The data used in this work is openly licensed as S1, S2, and the Copernicus EMS flood products are freely available under the Copernicus open-data policy (Copernicus, n.d.). This supports reproducibility and is also a fairness consideration, as it allows under-resourced agencies to access the same observations as well-funded ones. However, there is an asymmetric cost to errors in the intended application of this work, where a false negative may leave affected populations without aid, whereas a false positive is comparatively less harmful. The per-sample analysis showed all three models failing on low-coverage and narrow riverine floods, precisely the cases where missed detections could have the greatest human cost, so outputs should support rather than replace human decisions. The equity of coverage is another concern, as the optical modality may perform poorly under the cloud cover that frequently accompanies severe flooding. Finally, there is the environmental cost of GPU acceleration during a hyper-parameter grid search. This motivated the use of a compact ResNet-34 encoder over heavier alternatives, consistent with evidence that larger encoders did not substantially improve performance (Ghosh et al., 2024).

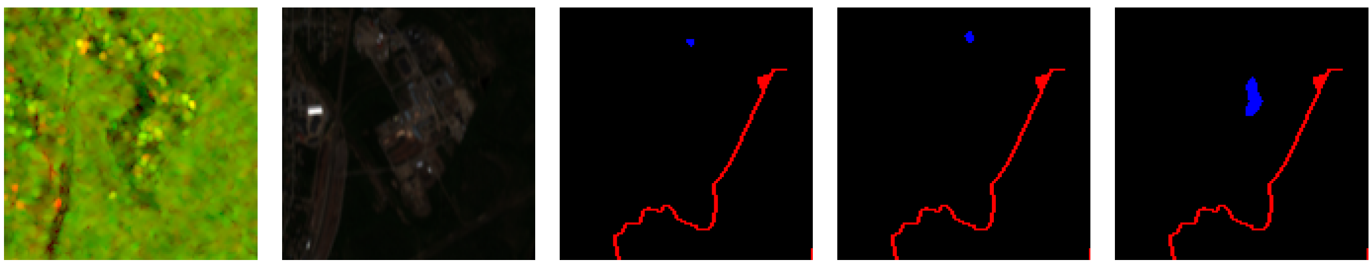
VI. Conclusion

This project set out to evaluate whether multimodal fusion of S1 SAR and S2 optical imagery improves flood extent segmentation compared to single-modality approaches. To assess this, a feature-fusion model was compared against S1 and S2 baselines. All three models used the same U-Net architecture and ResNet-34 encoder, trained and evaluated on the STURM-Fusion dataset. Each model was selected via the same grid

(a) Input Image: SAR (b) Input Image: Optical (c) FN/FP Map: Fusion (d) FN/FP Map: Optical (e) FN/FP Map: SAR

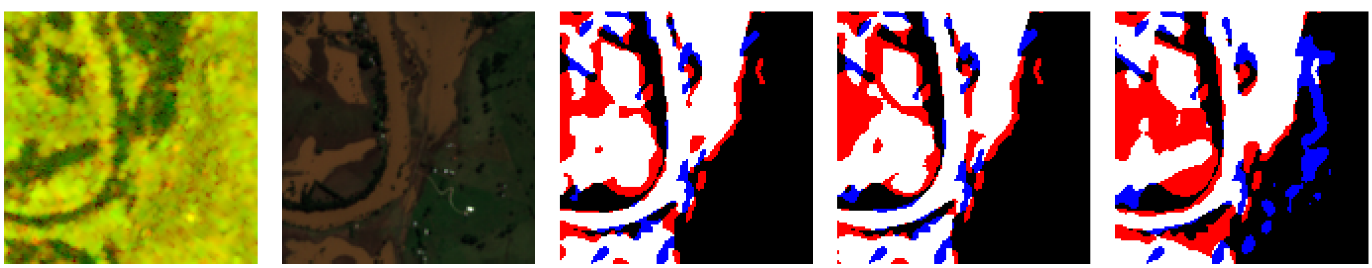
Fig. 3: EMSR470_AOI01_46_07_2_1: Fusion(IoU 0.9684), Optical(IoU 0.8969), SAR(IoU 0.9745).

FN/FP maps: black = correct background, white = correct flood, red = false negative, blue = false positive.



(a) Input Image: SAR (b) Input Image: Optical (c) FN/FP Map: Fusion (d) FN/FP Map: Optical (e) FN/FP Map: SAR

Fig. 4: EMSR441_AOI05_2_3_2_2: Fusion(IoU 0.0), Optical(IoU 0.0), SAR(IoU 0.0)



(a) Input Image: SAR (b) Input Image: Optical (c) FN/FP Map: Fusion (d) FN/FP Map: Optical (e) FN/FP Map: SAR

Fig. 5: EMSR570_AOI02_07_03_2_1: Fusion(IoU 0.6983), Optical(IoU 0.6693), SAR(IoU 0.5619)

search procedure and assessed on a held-out test set, with IoU as the primary segmentation metric, along with supporting per-sample and qualitative analyses. The results showed that the fusion model was the strongest performer overall, achieving the highest aggregate IoU of the three models. However, while there was a significant increase in performance over the S1 model, there was only a marginal increase over the S2 optical baseline. This pattern was clear in the qualitative examples and confirmed by the cross-model improvement distribution, where fusion matched the best available single-modality rather than exceeding it. Taken together, these findings indicate that the behaviour of the fusion model was closely tied to the stronger modality, most often the optical one. The fusion model did not, under these conditions, demonstrate a clear benefit from combining the two modalities. The results, however, should be considered in light of the dataset's limitations. STURM-Fusion inherits a preprocessing pipeline that filters out the most cloud-affected tiles, under-representing the conditions in which SAR is expected to compensate for optical imagery. Addressing these limitations suggests several directions for future work. The clearest next step is to construct a dataset anchored on the flood mask rather than on a preprocessed optical dataset, selecting tiles to balance flood coverage and matching imagery under a more permissive cloud threshold. This would test fusion's ability to compensate for degraded optical imagery directly, rather than removing those conditions before evaluation. A second step would be to evaluate the model's ability to generalise to unseen flood events by implementing an event-based or location-based split. Finally, a longer-term direction is real-time flood mapping. Copernicus EMS already produces flood maps from S1 and S2 during emergencies, and a fusion model could assist in automating this process.

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Appendix A
Dataset

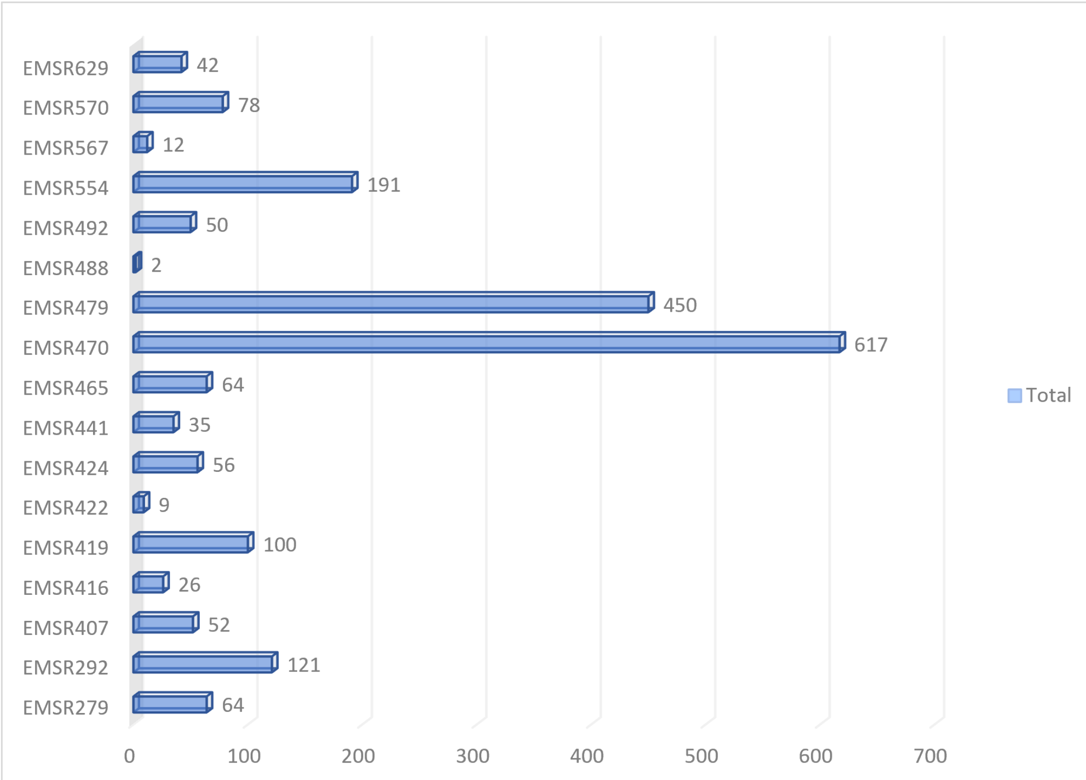


Fig. 6: Number of samples per EMSR event.

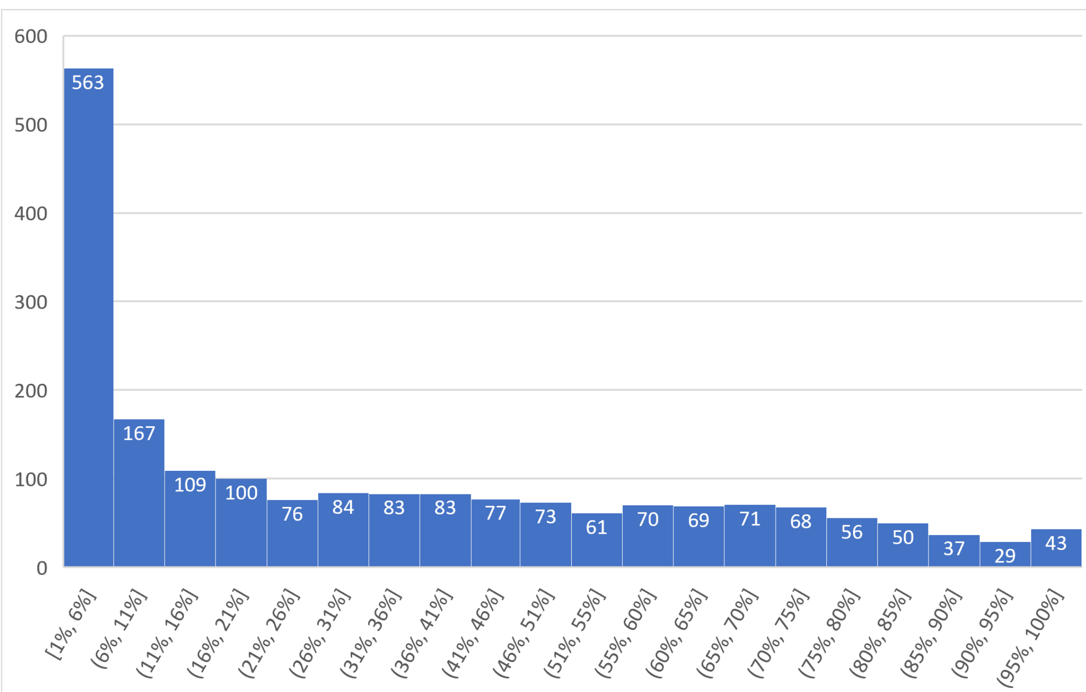


Fig. 7: Distribution of flood coverage (% of pixels) per tile.

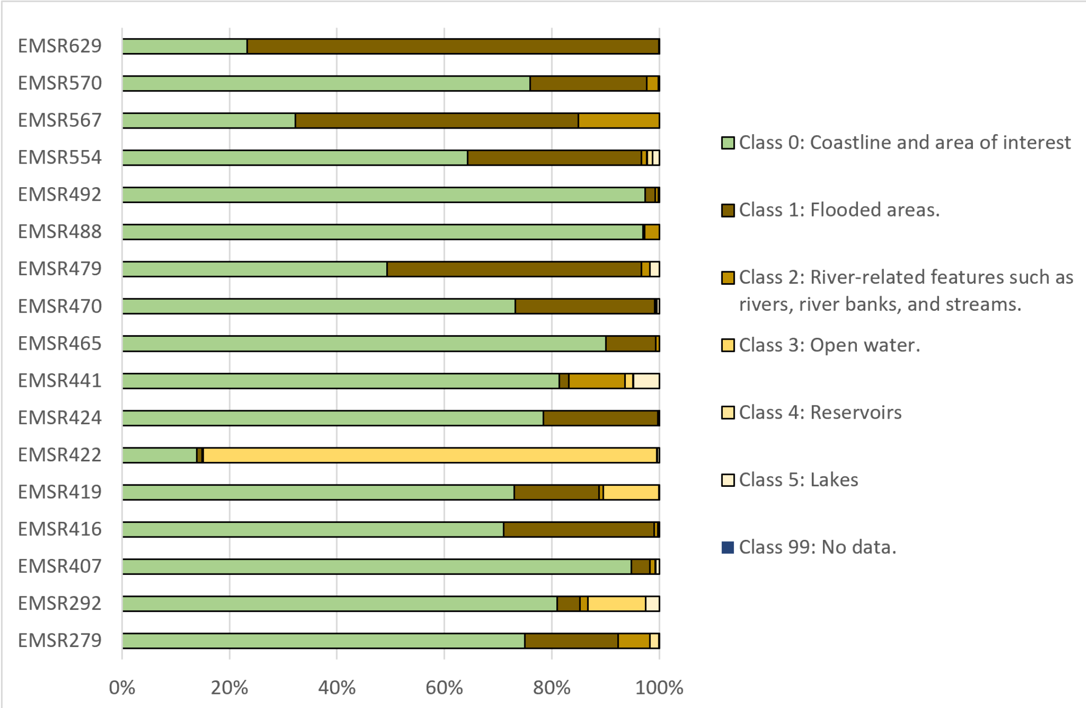


Fig. 8: Class percentage per EMSR event from the original multi-class labels.

Appendix B
Model Architecture

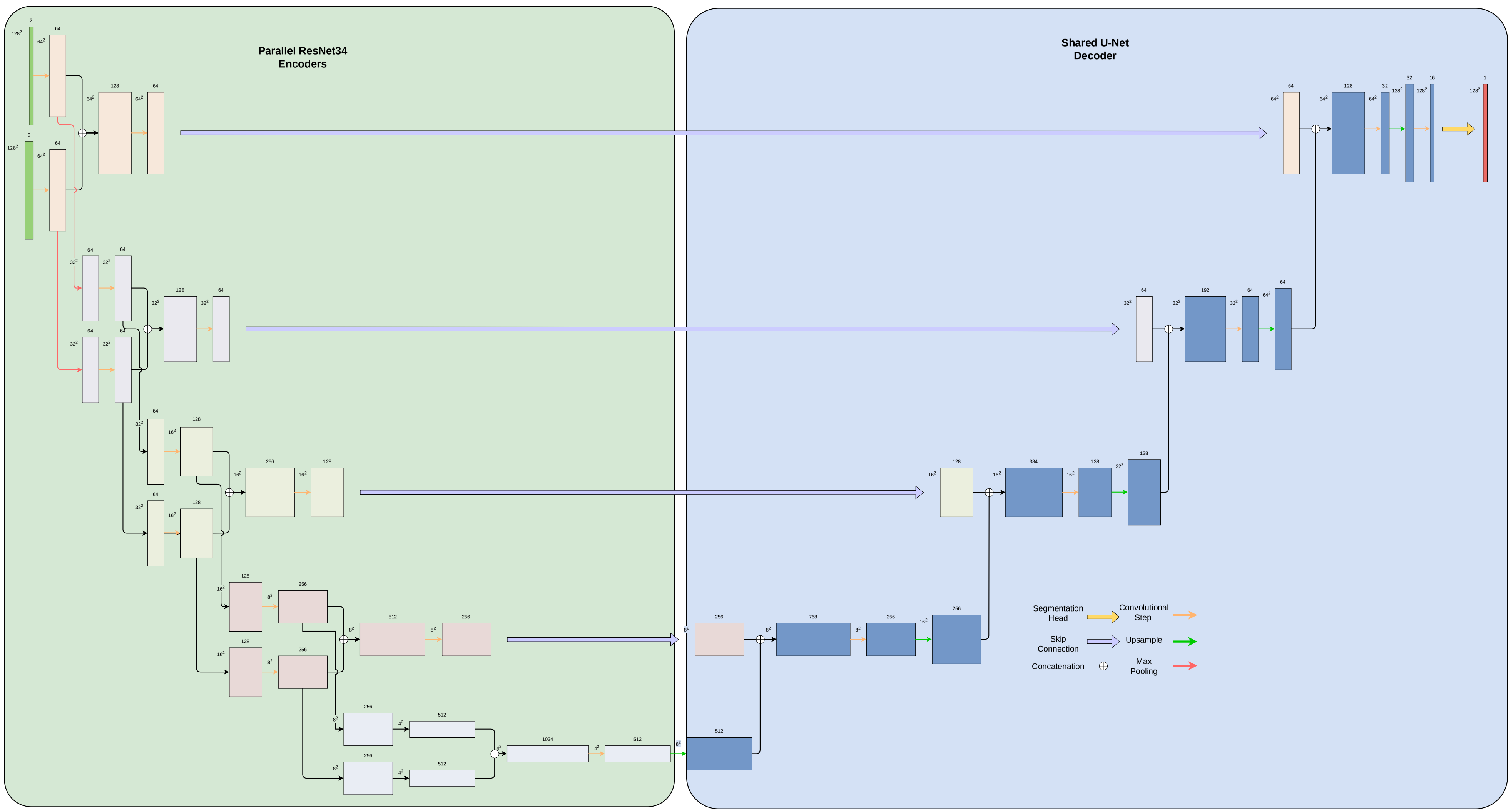
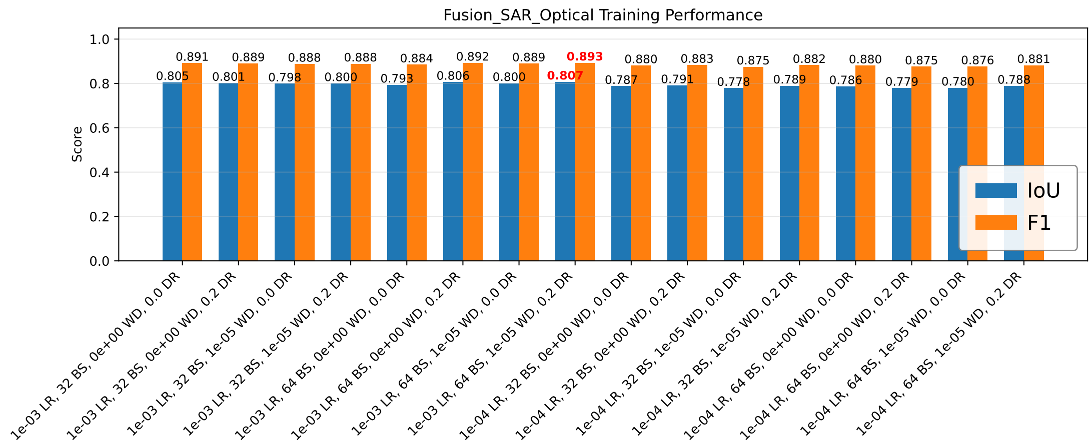


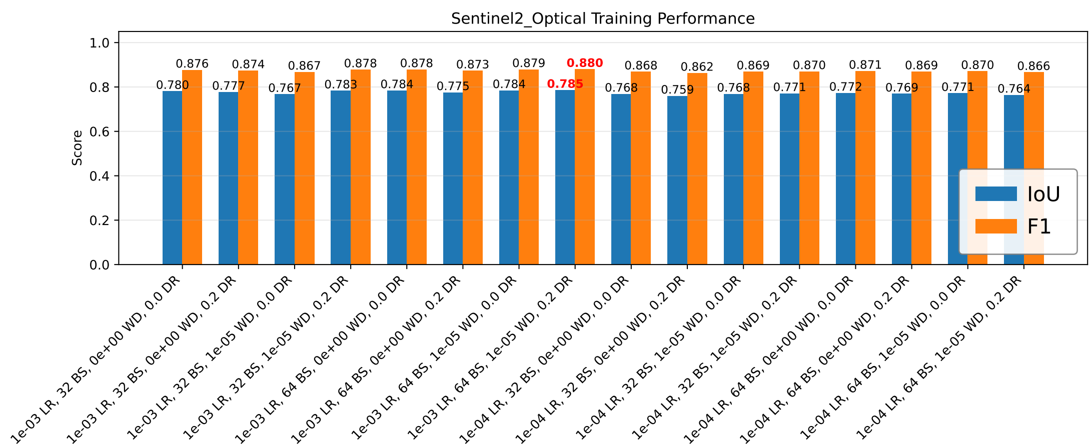
Fig. 9: Full feature-fusion parallel-encoder and decoder architecture.

APPENDIX C
MODEL TRAINING

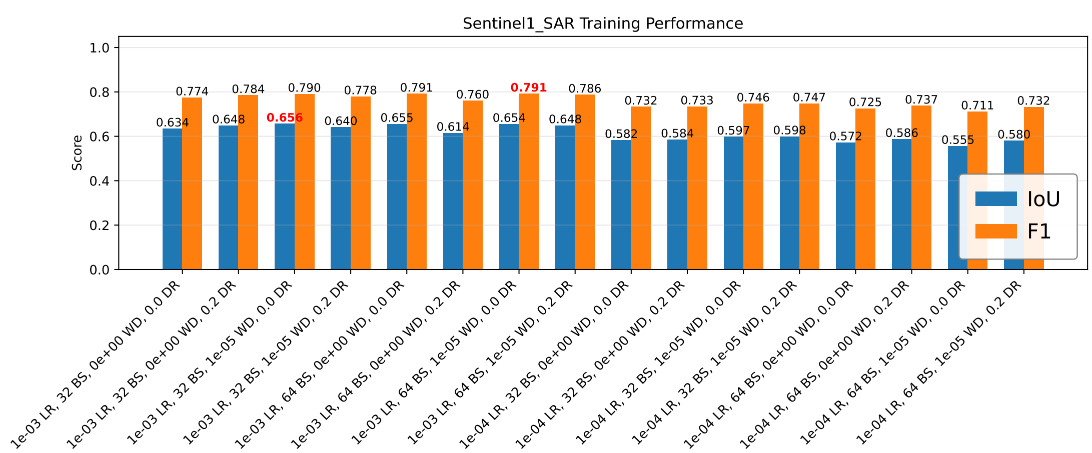
A. Hyperparameter Tuning Results



(a) Fusion model.



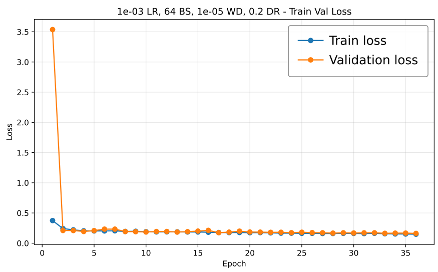
(b) Sentinel-2 (optical) model.



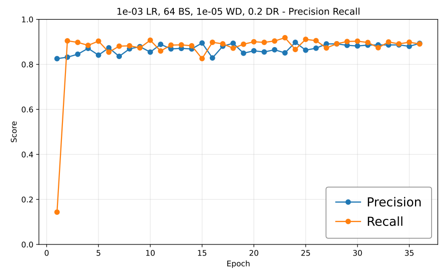
(c) Sentinel-1 (SAR) model.

Fig. 10: IoU and F1 score for all models trained during hyperparameter tuning, grouped by model type.

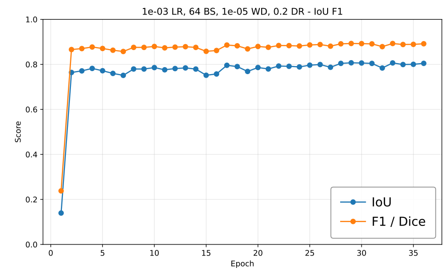
B. Training Curves



(a) Training and validation loss.

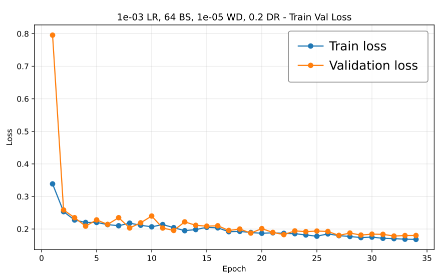


(b) Precision and recall.

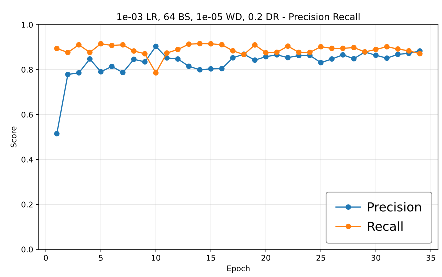


(c) IoU and F1 score.

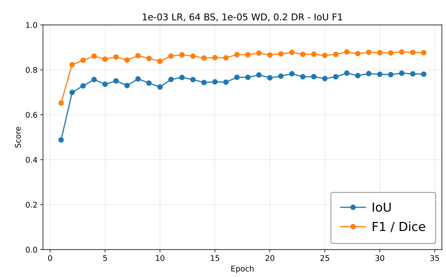
Fig. 11: Training curves for the fusion model.



(a) Training and validation loss.

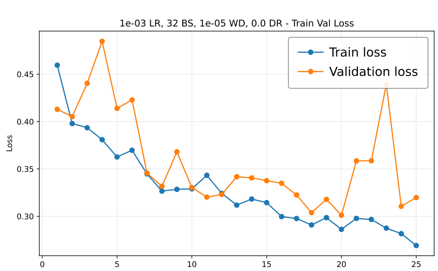


(b) Precision and recall.

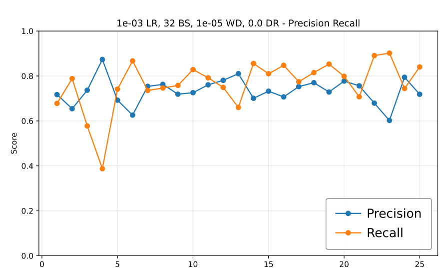


(c) IoU and F1 score.

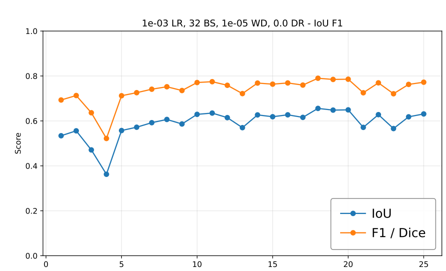
Fig. 12: Training curves for the Sentinel-2 (optical) model.



(a) Training and validation loss.



(b) Precision and recall.

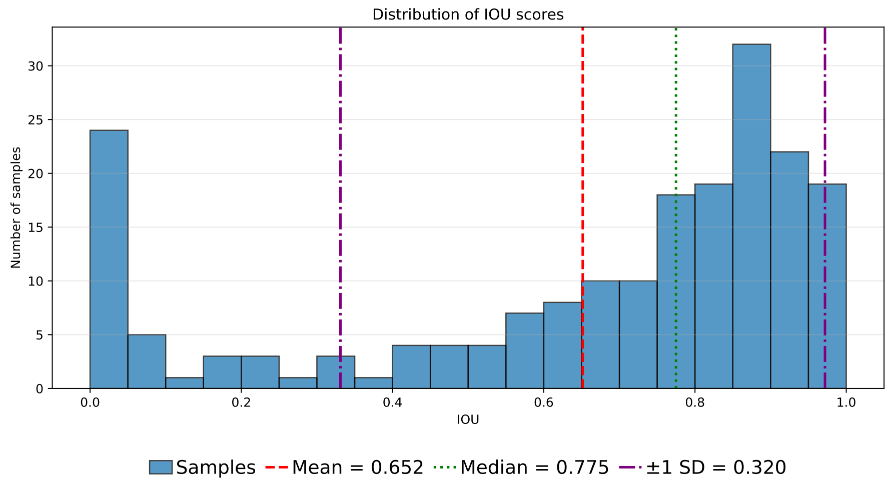


(c) IoU and F1 score.

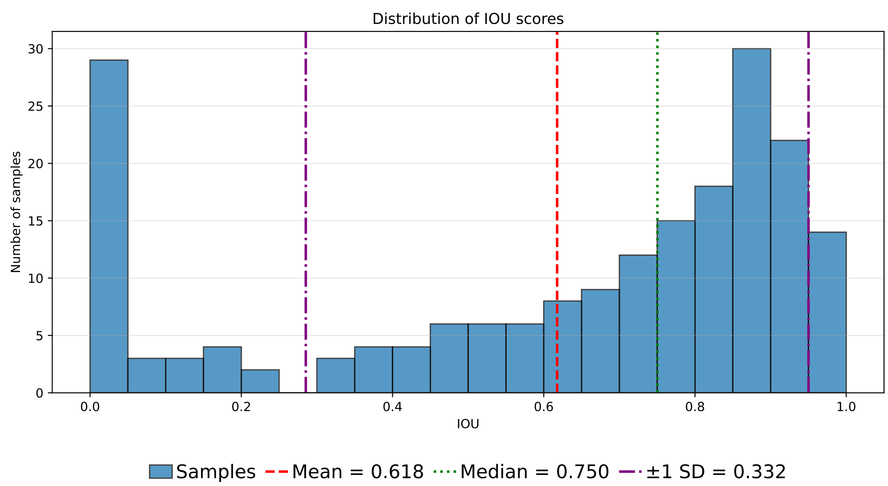
Fig. 13: Training curves for the Sentinel-1 (SAR) model.

APPENDIX D
EVALUATION RESULTS

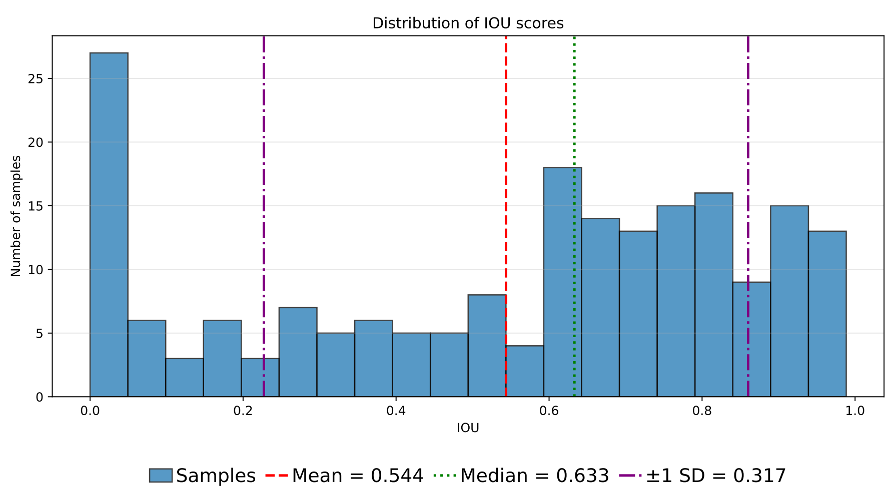
A. IoU Distributions



(a) Fusion model.



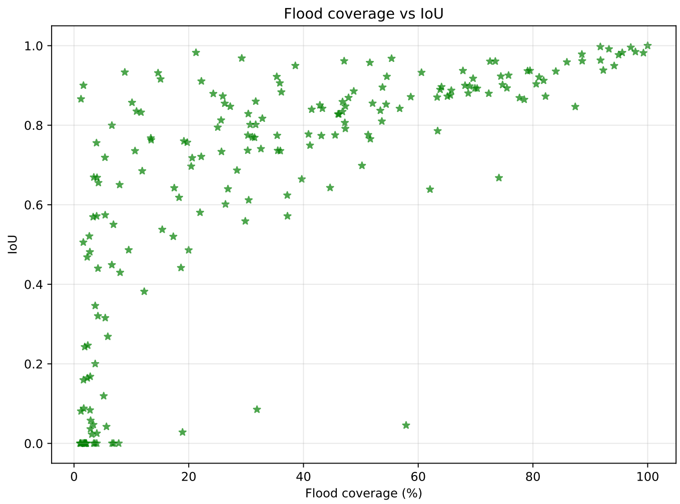
(b) Sentinel-2 (optical).



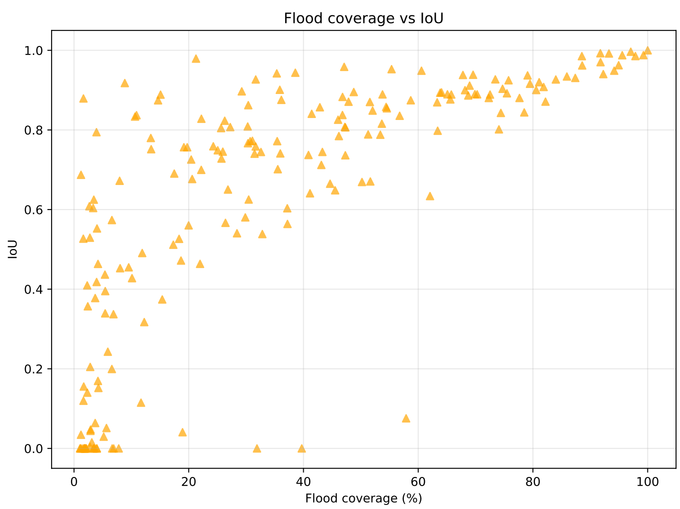
(c) Sentinel-1 (SAR).

Fig. 14: Distribution of IoU scores by model type.

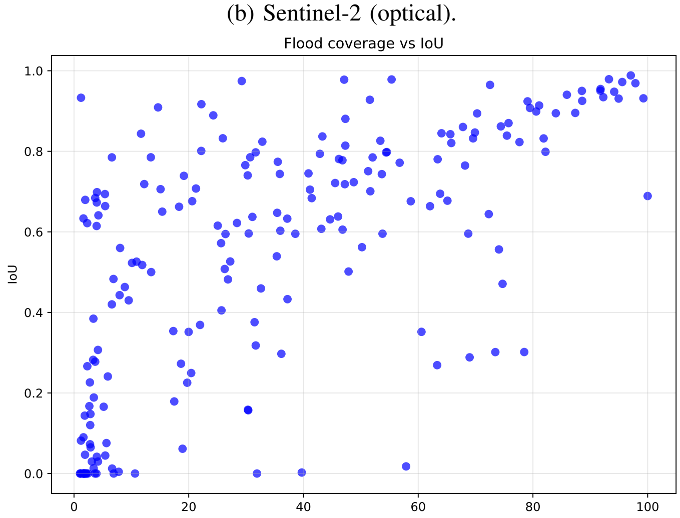
B. Scatter Plots



(a) Fusion model.



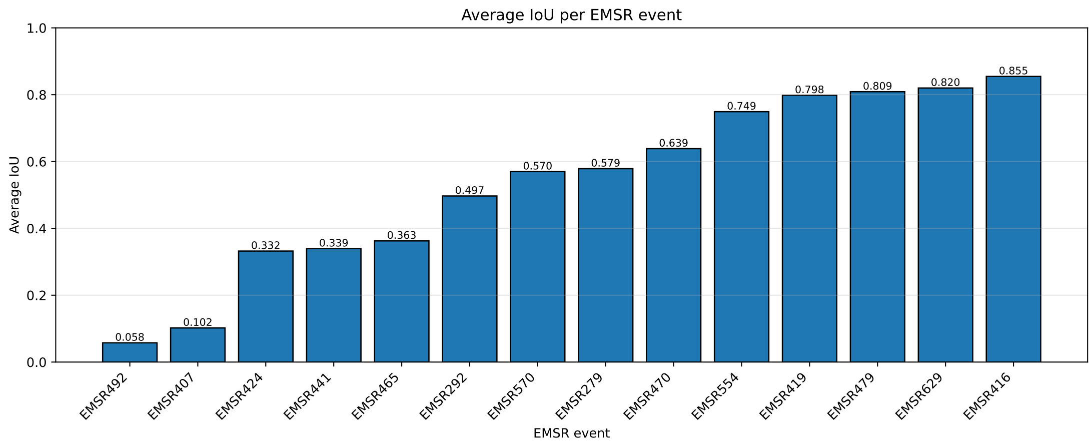
(b) Sentinel-2 (optical).



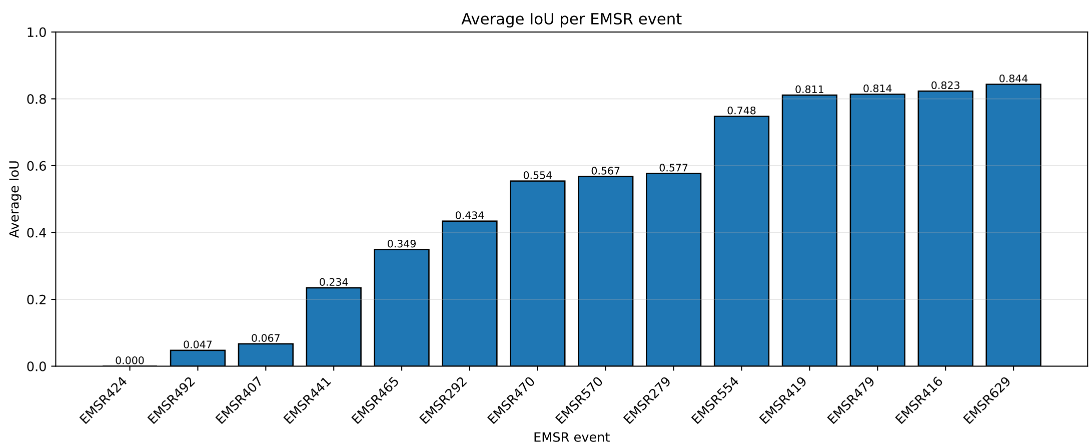
(c) Sentinel-1 (SAR).

Fig. 15: Scatter plots for each model type.

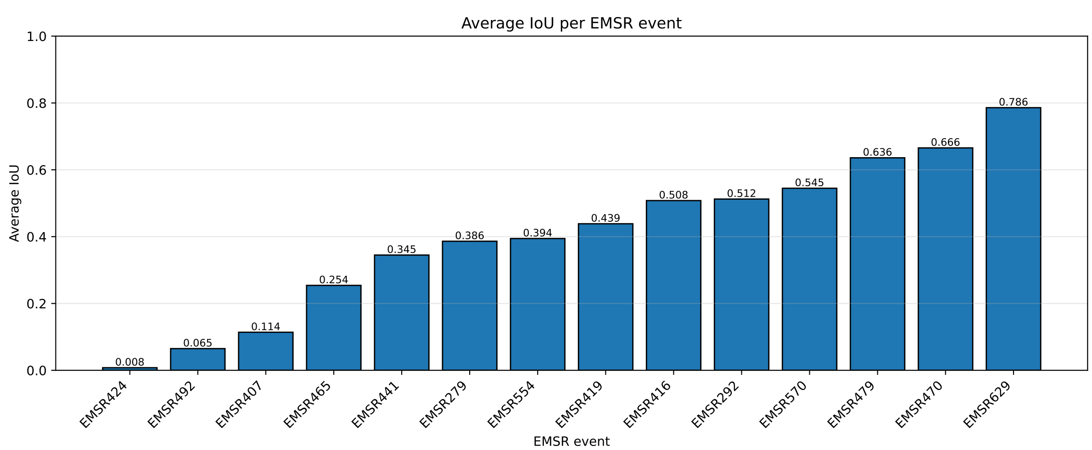
C. Average IoU per Event



(a) Fusion model.



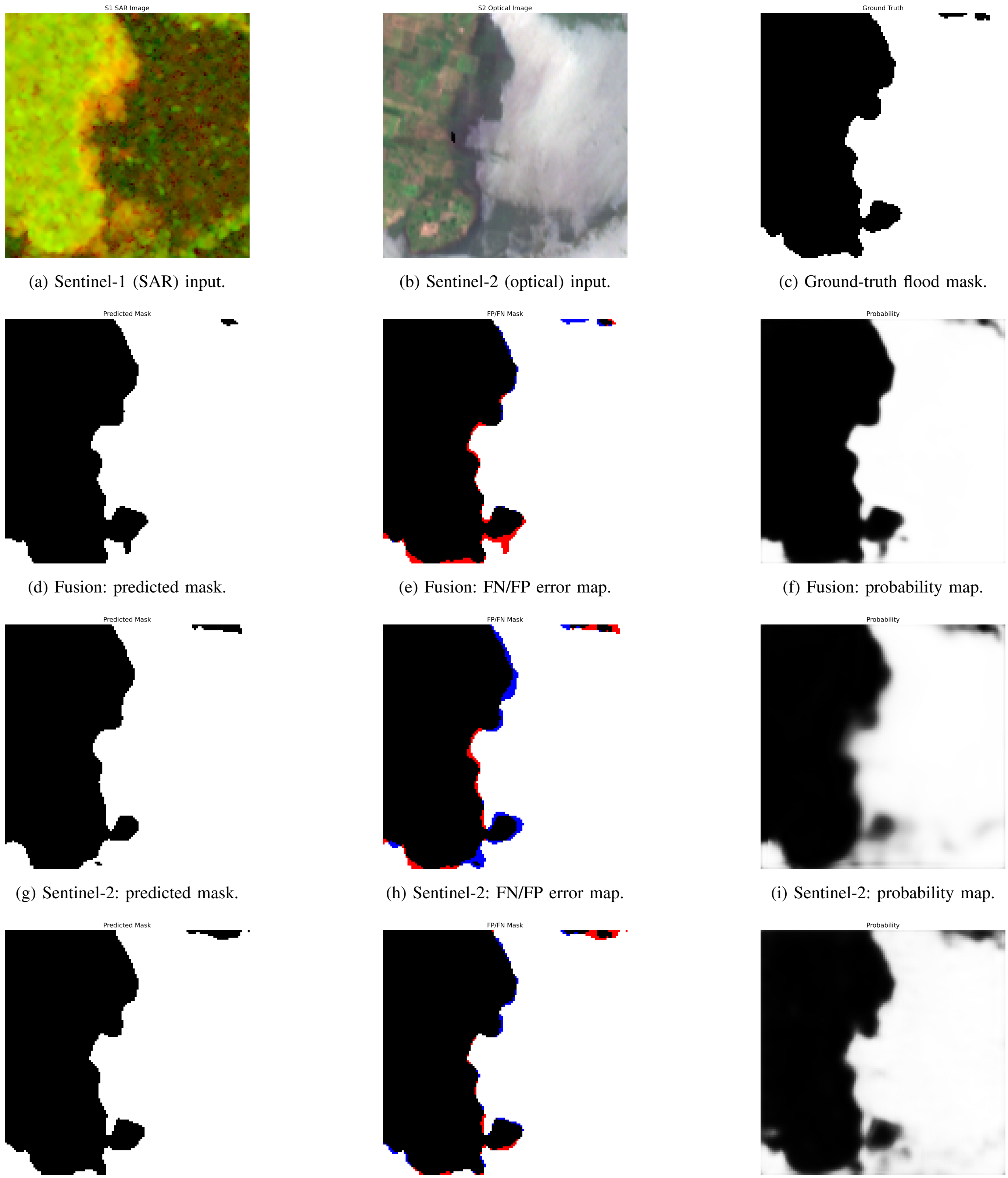
(b) Sentinel-2 (optical).



(c) Sentinel-1 (SAR).

Fig. 16: Average IoU per EMSR event for each model type.

D. Additional Examples



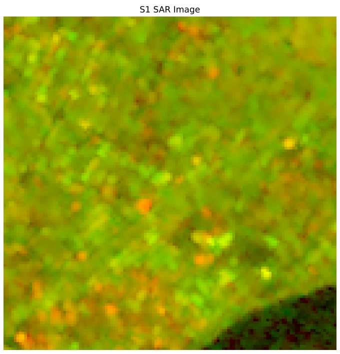
(a) Sentinel-1 (SAR) input. (b) Sentinel-2 (optical) input. (c) Ground-truth flood mask.

(d) Fusion: predicted mask. (e) Fusion: FN/FP error map. (f) Fusion: probability map.

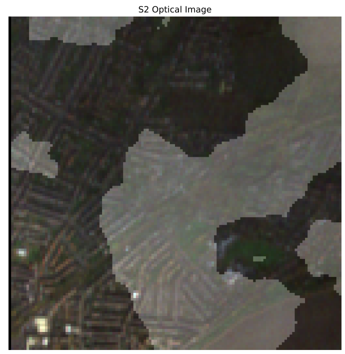
(g) Sentinel-2: predicted mask. (h) Sentinel-2: FN/FP error map. (i) Sentinel-2: probability map.

(j) Sentinel-1: predicted mask. (k) Sentinel-1: FN/FP error map. (l) Sentinel-1: probability map.

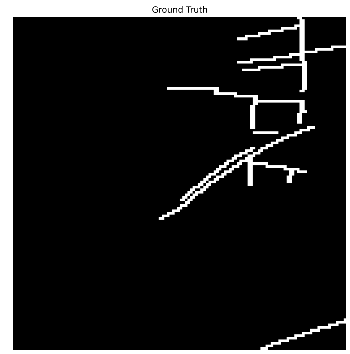
Fig. 17: EMSR470_AOI01_29_13_2_2: Fusion (IoU 0.9677), Optical (IoU 0.9527), SAR (IoU 0.9784). A success case with a clear boundary and just over half the tile flooded. Despite possible cloud cover in the optical input (b), all models perform well, with errors confined to the boundary. SAR scores highest and fusion outperforms optical, supporting SAR's contribution under cloud.



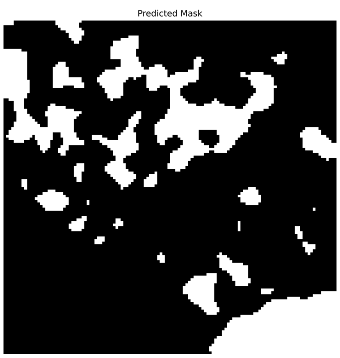
(a) Sentinel-1 (SAR) input.



(b) Sentinel-2 (optical) input.



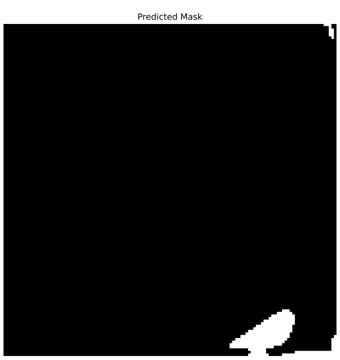
(c) Ground-truth flood mask.



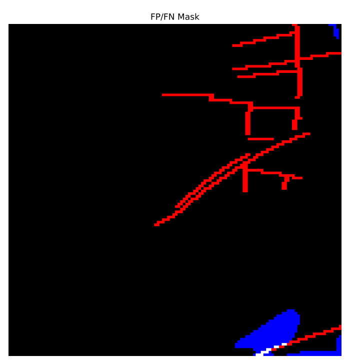
(d) Fusion: predicted mask.

(e) Fusion: FN/FP error map.

(f) Fusion: probability map.

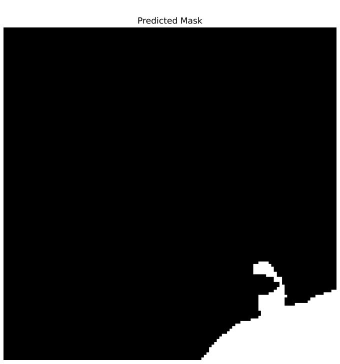


(g) Sentinel-2: predicted mask.

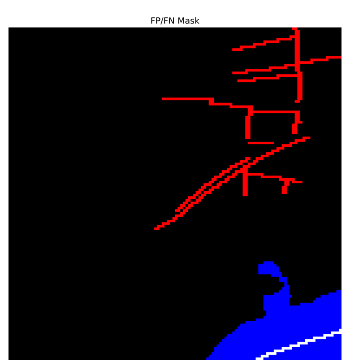


(h) Sentinel-2: FN/FP error map.

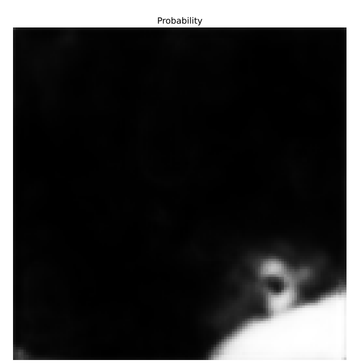
(i) Sentinel-2: probability map.



(j) Sentinel-1: predicted mask.

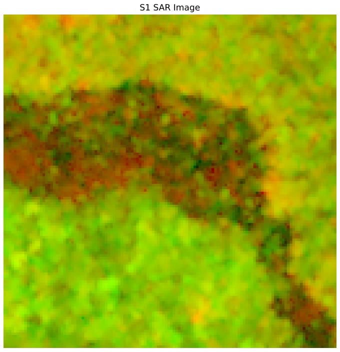


(k) Sentinel-1: FN/FP error map.

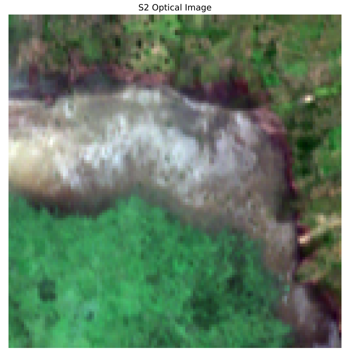


(l) Sentinel-1: probability map.

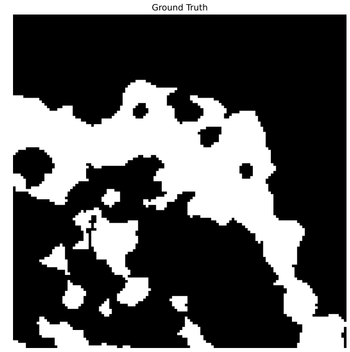
Fig. 18: EMSR407_AOI01_03_17_2_1: Fusion (IoU 0.0226), Optical (IoU 0.0152), SAR (IoU 0.0297). A near-total failure case for all three models. Cloud-cover effects similar to those in the success case appear to produce false positives across all models. SAR achieves the highest IoU and fusion follows it more closely than the optical model.



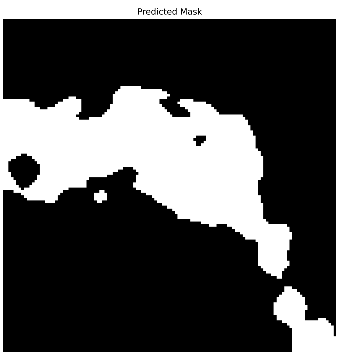
(a) Sentinel-1 (SAR) input.



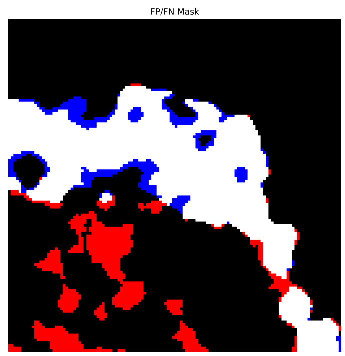
(b) Sentinel-2 (optical) input.



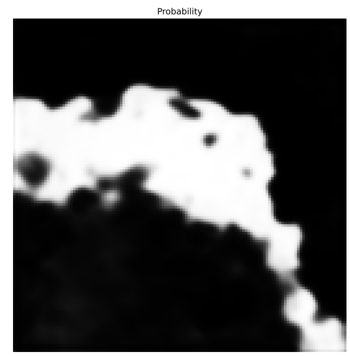
(c) Ground-truth flood mask.



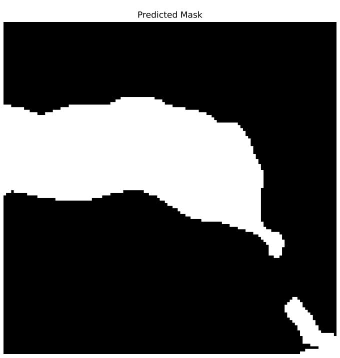
(d) Fusion: predicted mask.



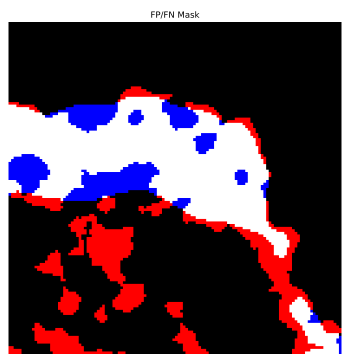
(e) Fusion: FN/FP error map.



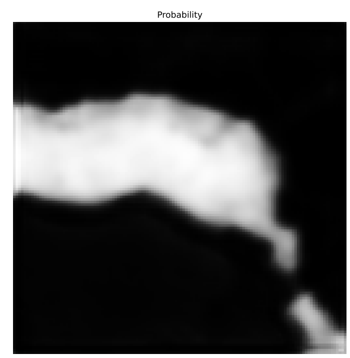
(f) Fusion: probability map.



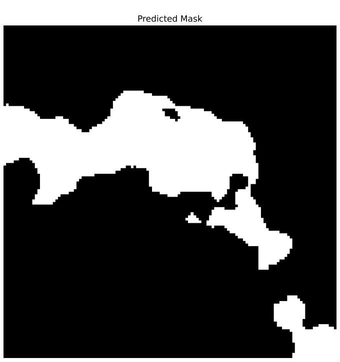
(g) Sentinel-2: predicted mask.



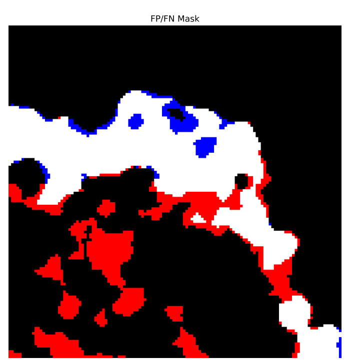
(h) Sentinel-2: FN/FP error map.



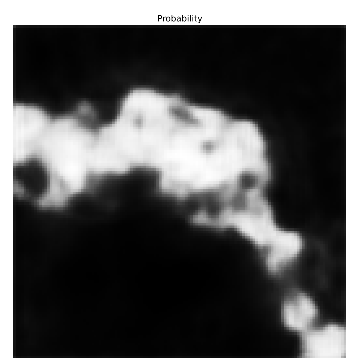
(i) Sentinel-2: probability map.



(j) Sentinel-1: predicted mask.

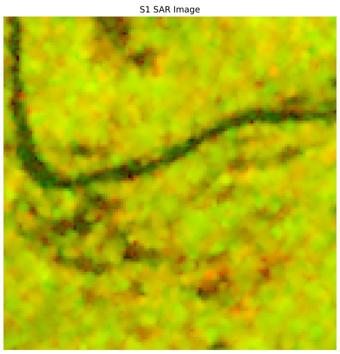


(k) Sentinel-1: FN/FP error map.

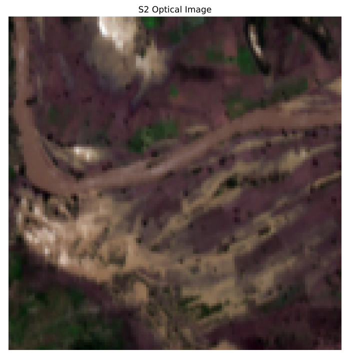


(l) Sentinel-1: probability map.

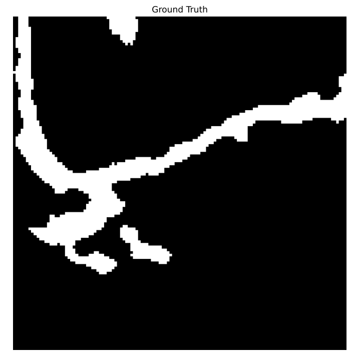
Fig. 19: EMSR470_AOI01_10_25_1_1: Fusion (IoU 0.6866), Optical (IoU 0.5404), SAR (IoU 0.6220). A false-negative-dominated case: all models under-predict the flood, but fusion recovers the most and SAR outperforms optical, supporting the benefit of SAR information in the fusion model.



(a) Sentinel-1 (SAR) input.

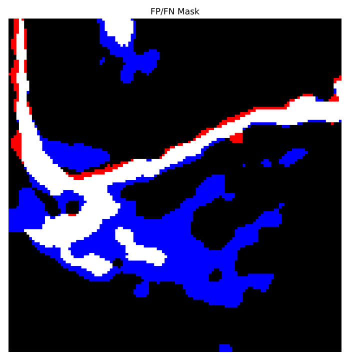


(b) Sentinel-2 (optical) input.



(c) Ground-truth flood mask.

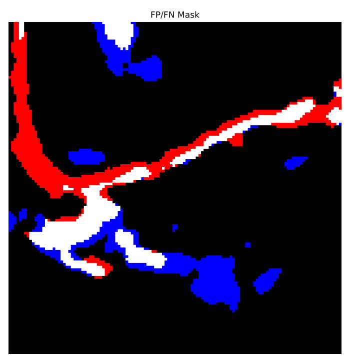
(d) Fusion: predicted mask.



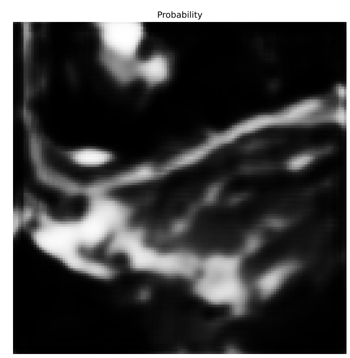
(e) Fusion: FN/FP error map.

(f) Fusion: probability map.

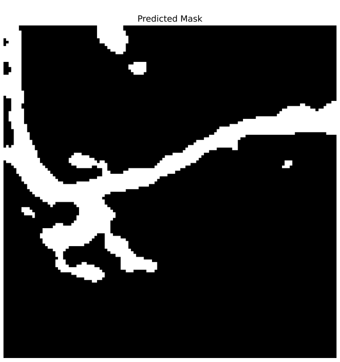
(g) Sentinel-2: predicted mask.



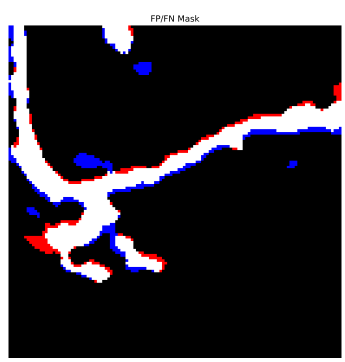
(h) Sentinel-2: FN/FP error map.



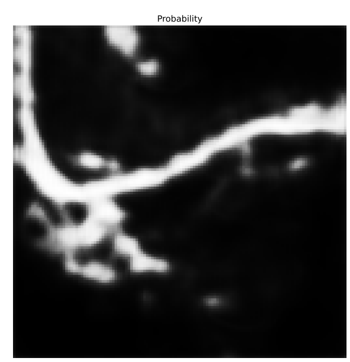
(i) Sentinel-2: probability map.



(j) Sentinel-1: predicted mask.

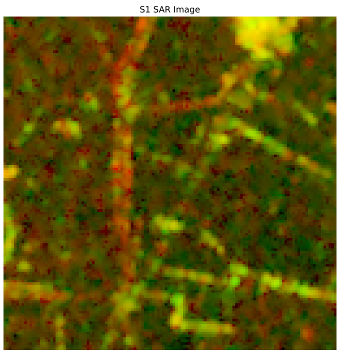


(k) Sentinel-1: FN/FP error map.



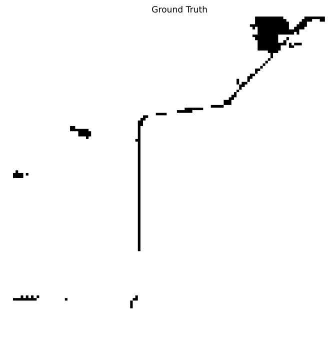
(l) Sentinel-1: probability map.

Fig. 20: EMSR470_AOI01_47_10_1_2: Fusion (IoU 0.3818), Optical (IoU 0.3173), SAR (IoU 0.7187). SAR substantially outperforms both other models, cleanly capturing the narrow channel. Fusion improves slightly on optical, showing some SAR benefit, but the strong optical signal limits the gain and fusion falls well short of SAR.

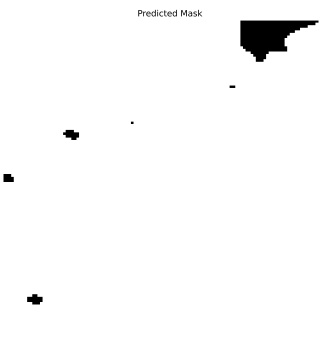


(a) Sentinel-1 (SAR) input.

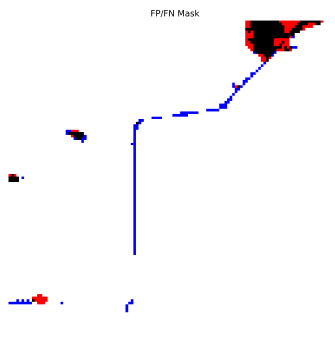
(b) Sentinel-2 (optical) input.



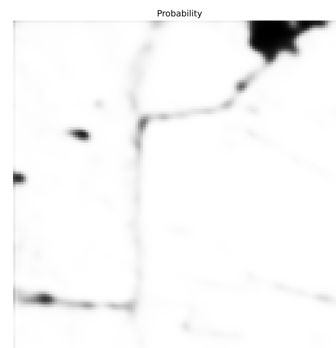
(c) Ground-truth flood mask.



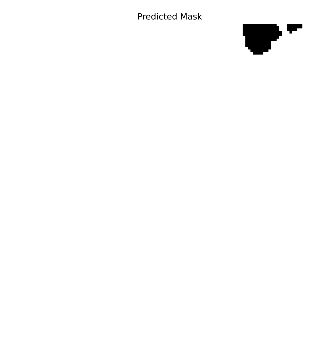
(d) Fusion: predicted mask.



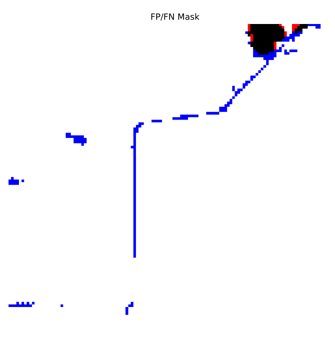
(e) Fusion: FN/FP error map.



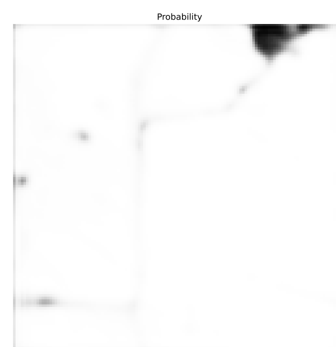
(f) Fusion: probability map.



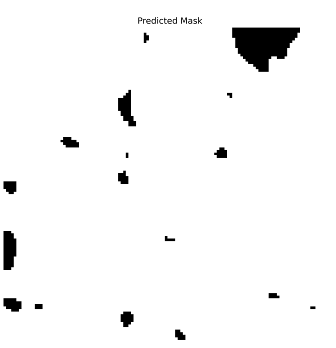
(g) Sentinel-2: predicted mask.



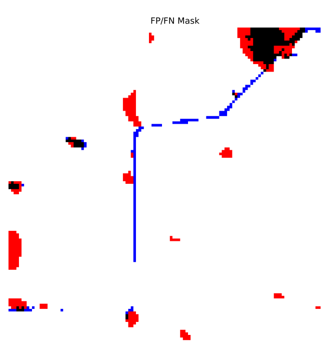
(h) Sentinel-2: FN/FP error map.



(i) Sentinel-2: probability map.



(j) Sentinel-1: predicted mask.



(k) Sentinel-1: FN/FP error map.

(l) Sentinel-1: probability map.

Fig. 21: EMSR629_AOI01_09_01_1_2: Fusion (IoU 0.9845), Optical (IoU 0.9855), SAR (IoU 0.9691). All models segment the dominant flooded region well but miss the thin channel. The probability maps (f, i, l) show the channel is detected, particularly by fusion, but falls below the 0.5 threshold and is lost in the binary masks indicating a thresholding limitation rather than a failure to learn the signal.

E. Fusion Improvement Distribution

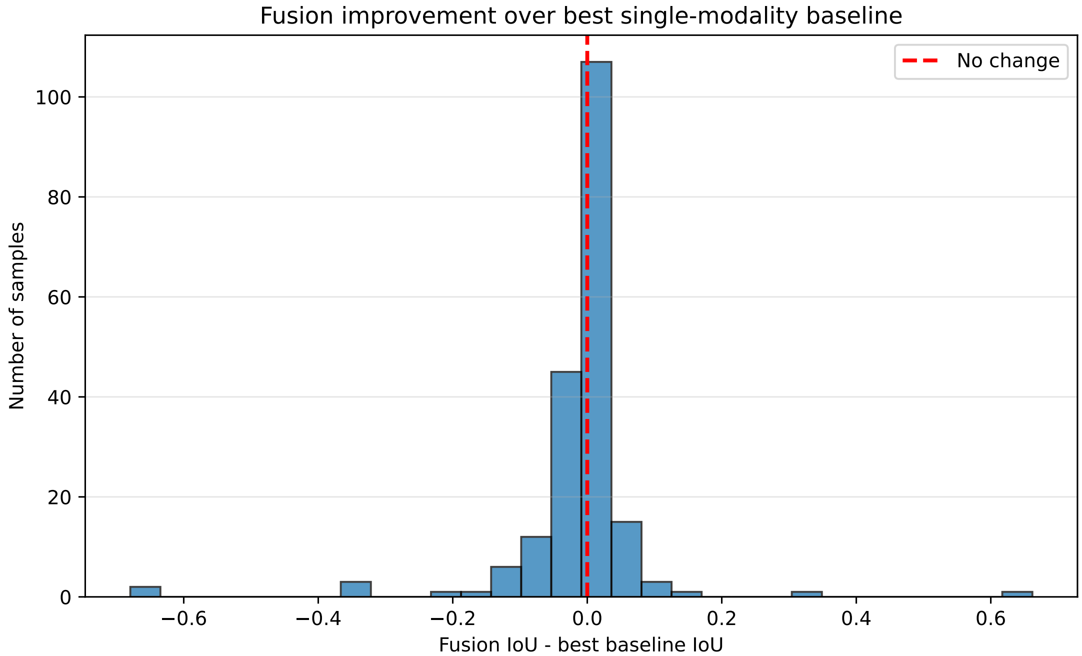


Fig. 22: Distribution of per-sample IoU improvement of the fusion model over the best single-modality baseline.

F. Generative AI Tools Student Statement

ECS750P/ECS753P/ECS754P– MSc Project Module
Generative AI Tools – Student Accountability Statement

The completed and signed statement must be included in the appendix of the final dissertation paper.

Student Name: Taylor Millin Reade
Project Title: Evaluation of Multi-Modal Fusion for Deep Learning-Based Flood Extent Mapping Using Sentinel-1 and Sentinel-2
Supervisor Name: Dr Miles Hansard

1. Have you used Generative AI tools for your project?
YES
If yes, please complete sections 2 and 3 and sign the declaration below.
If no, please sign the declaration below.

2. Please provide details of all the AI tools that you used in your project work.

AI Tool	Version	Publisher	Access URL
ChatGPT	3.5	OpenAI	https://chat.openai.com/
Claude	Claude Sonnet 5	Anthropic	https://claude.ai/
Grammarly	Grammarly (Premium)	Grammarly Inc.	https://www.grammarly.com/

3. If you have used Artificial Intelligence (AI) tools in any part of your project please include in table format how, where, and why these tools were used. This should include references to the relevant parts of your project work — for example:

- Specific chapters or sections of the report (e.g. literature review, methodology, requirements)
- Software development (e.g. AI-assisted code generation, debugging, documentation)
- Interview or survey materials (e.g. AI-generated or refined questions/prompts)
- Datasets (e.g. AI-assisted data cleaning, labelling, or synthesis)
- The showcase video (e.g. AI-assisted scripting, narration, or editing support)

Be specific and transparent about your use of AI tools, including both direct outputs (e.g. text, code, images) and indirect uses (e.g. idea generation, editing assistance). The goal is to ensure academic integrity and provide a clear record of how AI contributed to your work.

Fig. 23: Generative AI Tools Student Statement (page 1 of 2)

Example Table – please add your own

Project Section	Description	Justification
Throughout Project - Research	Used to explain unfamiliar concepts and technologies (API's, libraries, remote sensing/deep learning concepts).	Used to understand unfamiliar concepts and technologies. I applied the understanding myself in my own writing and implementation.
Throughout Project - Planning/Drafting	Used to structure and plan how I would write up sections of the report	Used as a writing aid to organize my own ideas.
Throughout Project - Grammer/Spelling	Used Grammarly to check grammar, spelling, and sentence clarity across the written report.	Used for language correction, not content generation or rewriting. All analysis, arguments, and final text are my own.

3. Generative AI Student Declaration

- I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.
- I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- I take full responsibility for the originality and integrity of my project submission.
- I have referenced all sources, including those suggested or produced with AI support.
- I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed: ____________________ Date: 19/08/2026

Fig. 24: Generative AI Tools Student Statement (page 2 of 2)